

Creation Date 11-Sep-2014

Revision Date 27-Sep-2023

Revision Number 9

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **Hydrazine hydrate, 80% (Hydrazine, 51%)**  
Cat No. : **209590000; 209592500; 209690010**  
CAS No **10217-52-4**  
Molecular Formula **H4 N2 . x H2 O**

Unique Formula Identifier (UFI) **RDKY-82PW-MX03-P4NT**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**Swiss distributor - Fisher Scientific AG**  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

E-mail address [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

#### Poison Centre - Emergency information services

**Ireland** : National Poisons Information Centre (NPIC) -  
**01 809 2166** (8am-10pm, 7 days a week)  
**Malta** : +356 2395 2000  
**Cyprus** : +357 2240 5611

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Based on available data, the classification criteria are not met

##### Health hazards

|                                    |                     |
|------------------------------------|---------------------|
| Acute oral toxicity                | Category 3 (H301)   |
| Acute dermal toxicity              | Category 3 (H311)   |
| Acute Inhalation Toxicity - Vapors | Category 2 (H330)   |
| Skin Corrosion/Irritation          | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation  | Category 1 (H318)   |
| Skin Sensitization                 | Category 1 (H317)   |
| Carcinogenicity                    | Category 1B (H350)  |

##### Environmental hazards

|                          |                   |
|--------------------------|-------------------|
| Acute aquatic toxicity   | Category 1 (H400) |
| Chronic aquatic toxicity | Category 1 (H410) |

Full text of Hazard Statements: see section 16

### 2.2. Label elements



Signal Word

Danger

#### **Hazard Statements**

H330 - Fatal if inhaled  
H314 - Causes severe skin burns and eye damage  
H317 - May cause an allergic skin reaction  
H350 - May cause cancer  
H410 - Very toxic to aquatic life with long lasting effects  
H301 + H311 - Toxic if swallowed or in contact with skin  
Combustible liquid

#### **Precautionary Statements**

P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P302 + P350 - IF ON SKIN: Gently wash with plenty of soap and water  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
P310 - Immediately call a POISON CENTER or doctor/physician

## Additional EU labelling

Restricted to professional users

### 2.3. Other hazards

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component           | CAS No     | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                                                                                                     |
|---------------------|------------|-------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrazine (hydrate) | 10217-52-4 |                   | 80       | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 2 (H330)<br>Skin Corr. 1B (H314)<br>Skin Sens. 1 (H317)<br>Eye Dam. 1 (H318)<br>Carc. 1B (H350)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410)                        |
| Hydrazine           | 302-01-2   | EEC No. 206-114-9 | -        | Flam. Liq. 3 (H226)<br>Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 2 (H330)<br>Skin Corr. 1B (H314)<br>Skin Sens. 1 (H317)<br>Eye Dam. 1 (H318)<br>Carc. 1B (H350)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) |
| Water               | 7732-18-5  | 231-791-2         | 20       | -                                                                                                                                                                                                                                     |

| Component | Specific concentration limits (SCL's)                                                                | M-Factor | Component notes |
|-----------|------------------------------------------------------------------------------------------------------|----------|-----------------|
| Hydrazine | Eye Irrit. 2 (H319) :: 3%≤C<10%<br>Skin Corr. 1B (H314) :: C≥10%<br>Skin Irrit. 2 (H315) :: 3%≤C<10% | 10       | -               |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | advice.                                                                                                                                                                                                                                                                                                                        |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

## **4.2. Most important symptoms and effects, both acute and delayed**

Causes burns by all exposure routes. Difficulty in breathing. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

## **4.3. Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **5.1. Extinguishing media**

#### **Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Combustible material. Containers may explode when heated. Do not allow run-off from fire-fighting to enter drains or water courses.

#### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Ammonia, Hydrogen.

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

## 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

## 6.3. Methods and material for containment and cleaning up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition.

## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition.

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep away from heat, sparks and flame. Store under an inert atmosphere. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1A

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### **Exposure limits**

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

| Component | European Union                                                                                                                                            | The United Kingdom                                                                                                                              | France                                                                                                                             | Belgium                                                                                  | Spain                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrazine | TWA: 0.013 mg/m <sup>3</sup> (8h)<br>TWA: 0.01 ppm (8h)<br>Skin                                                                                           | STEL: 0.03 ppm 15 min<br>STEL: 0.039 mg/m <sup>3</sup> 15 min<br>TWA: 0.01 ppm 8 hr<br>TWA: 0.013 mg/m <sup>3</sup> 8 hr<br>Carc.<br>Skin       | TWA / VME: 0.01 ppm (8 heures).<br>TWA / VME: 0.013 mg/m <sup>3</sup> (8 heures).<br>Peau                                          | TWA: 0.01 ppm 8 uren<br>TWA: 0.013 mg/m <sup>3</sup> 8 uren<br>Huid                      | TWA / VLA-ED: 0.01 ppm (8 horas)<br>TWA / VLA-ED: 0.013 mg/m <sup>3</sup> (8 horas)<br>Piel                                                                                       |
| Component | Italy                                                                                                                                                     | Germany                                                                                                                                         | Portugal                                                                                                                           | The Netherlands                                                                          | Finland                                                                                                                                                                           |
| Hydrazine | TWA: 0.013 mg/m <sup>3</sup> 8 ore. Time Weighted Average<br>TWA: 0.01 ppm 8 ore. Time Weighted Average<br>Pelle                                          | Haut                                                                                                                                            | TWA: 0.01 ppm 8 horas<br>TWA: 0.013 mg/m <sup>3</sup> 8 horas<br>Pele                                                              | huid<br>TWA: 0.013 mg/m <sup>3</sup> 8 uren                                              | TWA: 0.01 ppm 8 tunteina<br>TWA: 0.013 mg/m <sup>3</sup> 8 tunteina<br>STEL: 0.05 ppm 15 minuutteina<br>STEL: 0.07 mg/m <sup>3</sup> 15 minuutteina<br>Iho                        |
| Component | Austria                                                                                                                                                   | Denmark                                                                                                                                         | Switzerland                                                                                                                        | Poland                                                                                   | Norway                                                                                                                                                                            |
| Hydrazine | TRK-KZGW: 0.04 ppm 15 Minuten<br>TRK-KZGW: 0.052 mg/m <sup>3</sup> 15 Minuten<br>Haut<br>TRK-TMW: 0.01 ppm<br>TRK-TMW: 0.013 mg/m <sup>3</sup>            | TWA: 0.01 ppm 8 timer<br>TWA: 0.013 mg/m <sup>3</sup> 8 timer<br>STEL: 0.02 ppm 15 minutter<br>STEL: 0.026 mg/m <sup>3</sup> 15 minutter<br>Hud | Haut/Peau<br>TWA: 0.01 ppm 8 Stunden<br>TWA: 0.013 mg/m <sup>3</sup> 8 Stunden                                                     | STEL: 0.039 mg/m <sup>3</sup> 15 minutach<br>TWA: 0.013 mg/m <sup>3</sup> 8 godzinach    | TWA: 0.01 ppm 8 timer<br>TWA: 0.01 mg/m <sup>3</sup> 8 timer<br>STEL: 0.03 ppm 15 minutter. value calculated<br>STEL: 0.03 mg/m <sup>3</sup> 15 minutter. value calculated<br>Hud |
| Component | Bulgaria                                                                                                                                                  | Croatia                                                                                                                                         | Ireland                                                                                                                            | Cyprus                                                                                   | Czech Republic                                                                                                                                                                    |
| Hydrazine | TWA: 0.013 mg/m <sup>3</sup><br>TWA: 0.01 ppm<br>Skin notation                                                                                            | kože<br>TWA-GVI: 0.01 ppm 8 satima.<br>TWA-GVI: 0.013 mg/m <sup>3</sup> 8 satima.                                                               | TWA: 0.01 ppm 8 hr.<br>TWA: 0.013 mg/m <sup>3</sup> 8 hr.<br>STEL: 0.03 ppm 15 min<br>STEL: 0.039 mg/m <sup>3</sup> 15 min<br>Skin | Skin-potential for cutaneous absorption<br>TWA: 0.013 mg/m <sup>3</sup><br>TWA: 0.01 ppm | TWA: 0.05 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 0.025 mg/m <sup>3</sup>                                                                 |
| Component | Estonia                                                                                                                                                   | Gibraltar                                                                                                                                       | Greece                                                                                                                             | Hungary                                                                                  | Iceland                                                                                                                                                                           |
| Hydrazine | Nahk<br>TWA: 0.01 ppm 8 tundides.<br>TWA: 0.013 mg/m <sup>3</sup> 8 tundides.<br>STEL: 0.3 ppm 15 minutites.<br>STEL: 0.4 mg/m <sup>3</sup> 15 minutites. |                                                                                                                                                 | skin - potential for cutaneous absorption<br>TWA: 0.01 ppm<br>TWA: 0.013 mg/m <sup>3</sup>                                         | TWA: 0.013 mg/m <sup>3</sup> 8 óraban. AK<br>lehetséges borön keresztül felszívódás      | TWA: 0.01 ppm 8 klukkustundum.<br>TWA: 0.013 mg/m <sup>3</sup> 8 klukkustundum.<br>Skin notation<br>Ceiling: 0.2 ppm<br>Ceiling: 0.26 mg/m <sup>3</sup>                           |
| Component | Latvia                                                                                                                                                    | Lithuania                                                                                                                                       | Luxembourg                                                                                                                         | Malta                                                                                    | Romania                                                                                                                                                                           |
| Hydrazine | skin - potential for cutaneous exposure<br>TWA: 0.013 mg/m <sup>3</sup><br>TWA: 0.01 ppm                                                                  | TWA: 0.013 mg/m <sup>3</sup> IPRD<br>TWA: 0.01 ppm IPRD<br>Oda                                                                                  |                                                                                                                                    |                                                                                          | Skin notation<br>TWA: 0.08 ppm 8 ore<br>TWA: 0.1 mg/m <sup>3</sup> 8 ore<br>STEL: 0.8 ppm 15 minute<br>STEL: 1 mg/m <sup>3</sup> 15 minute                                        |
| Component | Russia                                                                                                                                                    | Slovak Republic                                                                                                                                 | Slovenia                                                                                                                           | Sweden                                                                                   | Turkey                                                                                                                                                                            |
| Hydrazine | TWA: 0.1 mg/m <sup>3</sup> 0528<br>Skin notation<br>MAC: 0.3 mg/m <sup>3</sup>                                                                            | TWA: 0.1 ppm 8 hodinách<br>TWA: 0.13 mg/m <sup>3</sup> 8 hodinách<br>Potential for cutaneous absorption                                         | TWA: 0.01 ppm 8 urah<br>TWA: 0.013 mg/m <sup>3</sup> 8 urah<br>Koža                                                                | TLV: 0.01 ppm 8 timmar. NGV<br>TLV: 0.013 mg/m <sup>3</sup> 8 timmar. NGV<br>Hud         |                                                                                                                                                                                   |

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

|  |  |                                                                       |  |  |  |
|--|--|-----------------------------------------------------------------------|--|--|--|
|  |  | STEL: 0.5 ppm 15 minútach<br>STEL: 0.65 mg/m <sup>3</sup> 15 minútach |  |  |  |
|--|--|-----------------------------------------------------------------------|--|--|--|

## Biological limit values

List source(s):

| Component | Italy | Finland | Denmark | Bulgaria | Romania                                              |
|-----------|-------|---------|---------|----------|------------------------------------------------------|
| Hydrazine |       |         |         |          | Hydrazine: 200 µg/g<br>Creatinine urine end of shift |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

No information available

## Predicted No Effect Concentration (PNEC)

No information available.

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

|                                 |                                                                                                                                                                                                                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                                                                               |
| Large scale/emergency use       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particulates filter conforming to EN 143 or Inorganic gases and vapours filter Type B Grey conforming to EN14387 |
| Small scale/Laboratory use      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Particle filtering: EN149:2001<br>When RPE is used a face piece Fit Test should be conducted                |
| Environmental exposure controls | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.                                                                                                                       |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                                                |                                          |
|-----------------------------------------|------------------------------------------------|------------------------------------------|
| Physical State                          | Liquid                                         |                                          |
| Appearance                              | Colorless                                      |                                          |
| Odor                                    | Ammonia-like                                   |                                          |
| Odor Threshold                          | No data available                              |                                          |
| Melting Point/Range                     | -57 °C / -70.6 °F                              |                                          |
| Softening Point                         | No data available                              |                                          |
| Boiling Point/Range                     | 117.2 °C / 243 °F                              |                                          |
| Flammability (liquid)                   | Combustible liquid                             | On basis of test data                    |
| Flammability (solid,gas)                | Not applicable                                 | Liquid                                   |
| Explosion Limits                        | <b>Lower</b> 4.8 Vol%<br><b>Upper</b> 100 Vol% |                                          |
| Flash Point                             | 91 °C / 195.8 °F                               | <b>Method -</b> No information available |
| Autoignition Temperature                | 310 °C / 590 °F                                |                                          |
| Decomposition Temperature               | No data available                              |                                          |
| pH                                      | 12                                             | 640 g/l aq.sol                           |
| Viscosity                               | 1.33 mPa.s at 20 °C                            |                                          |
| Water Solubility                        | Miscible                                       |                                          |
| Solubility in other solvents            | No information available                       |                                          |
| Partition Coefficient (n-octanol/water) |                                                |                                          |
| Component                               | <b>log Pow</b>                                 |                                          |
| Hydrazine                               | -0.16                                          |                                          |
| Vapor Pressure                          | 13 mbar @ 20 °C                                |                                          |
| Density / Specific Gravity              | 1.028                                          |                                          |
| Bulk Density                            | Not applicable                                 | Liquid                                   |
| Vapor Density                           | 1.1 @ 15 °C                                    | (Air = 1.0)                              |
| Particle characteristics                | Not applicable (liquid)                        |                                          |

### 9.2. Other information

|                      |                                        |
|----------------------|----------------------------------------|
| Molecular Formula    | H4 N2 . x H2 O                         |
| Molecular Weight     | 32.04                                  |
| Explosive Properties | explosive air/vapour mixtures possible |

## SECTION 10: STABILITY AND REACTIVITY

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

## 10.1. Reactivity

None known, based on information available

## 10.2. Chemical stability

Do not allow evaporation to dryness. Air sensitive.

## 10.3. Possibility of hazardous reactions

### Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
None under normal processing.

## 10.4. Conditions to avoid

Exposure to air. Incompatible products. Keep away from open flames, hot surfaces and sources of ignition.

## 10.5. Incompatible materials

Acids. Bases. Finely powdered metals. Halogens. nitrogen oxides (NOx). Organic materials. Peroxides. Lead. Metals. copper. Butyl rubber.

## 10.6. Hazardous decomposition products

Nitrogen oxides (NOx). Ammonia. Hydrogen.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 3 |
| Dermal     | Category 3 |
| Inhalation | Category 2 |

#### Toxicology data for the components

| Component | LD50 Oral               | LD50 Dermal                | LC50 Inhalation                              |
|-----------|-------------------------|----------------------------|----------------------------------------------|
| Hydrazine | LD50 = 60 mg/kg ( Rat ) | LD50 = 91 mg/kg ( Rabbit ) | 570 ppm ( Rat ) 4 h<br>0.75 mg/L ( Rat ) 4 h |
| Water     | -                       | -                          | -                                            |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

##### (d) respiratory or skin sensitization;

|             |                          |
|-------------|--------------------------|
| Respiratory | No data available        |
| Skin        | Category 1               |
|             | No information available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; Category 1B

May cause cancer The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU | UK | Germany | IARC |
|-----------|----|----|---------|------|
|-----------|----|----|---------|------|

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

|           |              |  |        |          |
|-----------|--------------|--|--------|----------|
| Hydrazine | Carc Cat. 1B |  | Cat. 2 | Group 2A |
|-----------|--------------|--|--------|----------|

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Fish                                                                                                                                                                                                                                                                                        | Water Flea | Freshwater Algae                                                                                                                                                                                 |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrazine | LC50: 0.28 - 1.34 mg/L, 96h static (Poecilia reticulata)<br>LC50: 1.81 - 2.79 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 1.17 mg/L, 96h (Lepomis macrochirus)<br>LC50: 0.54 - 1.31 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 0.7 - 1.3 mg/L, 96h flow-through (Lepomis macrochirus) |            | EC50: = 0.006 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: = 0.071 mg/L, 72h (Pseudokirchneriella subcapitata)<br>EC50: = 0.02 mg/L, 96h static (Pseudokirchneriella subcapitata) |

| Component           | Microtox                                                                     | M-Factor |
|---------------------|------------------------------------------------------------------------------|----------|
| Hydrazine (hydrate) | EC50 = 0.01 mg/L 15 min<br>EC50 = 0.01 mg/L 20 min<br>EC50 = 0.02 mg/L 5 min |          |
| Hydrazine           | EC50 = 0.01 mg/L 15 min<br>EC50 = 0.01 mg/L 20 min<br>EC50 = 0.02 mg/L 5 min | 10       |

### 12.2. Persistence and degradability

#### Persistence

Miscible with water, Persistence is unlikely, based on information available.

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

**Degradability**  
**Degradation in sewage treatment plant**

Not relevant for inorganic substances.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**12.3. Bioaccumulative potential**

Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Hydrazine | -0.16   | No data available             |

**12.4. Mobility in soil**

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

**12.5. Results of PBT and vPvB assessment**

No data available for assessment.

**12.6. Endocrine disrupting properties**

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

**12.7. Other adverse effects**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

**13.1. Waste treatment methods**

**Waste from Residues/Unused Products**

Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Solutions with high pH-value must be neutralized before discharge. Do not let this chemical enter the environment.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO**

**14.1. UN number**

UN2030

**14.2. UN proper shipping name**

HYDRAZINE, AQUEOUS SOLUTION

**14.3. Transport hazard class(es)**

8

**Subsidiary Hazard Class**

6.1

**14.4. Packing group**

II

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

## ADR

14.1. UN number UN2030  
 14.2. UN proper shipping name HYDRAZINE AQUEOUS SOLUTION  
 14.3. Transport hazard class(es) 8  
 Subsidiary Hazard Class 6.1  
 14.4. Packing group II

## IATA

14.1. UN number UN2030  
 14.2. UN proper shipping name HYDRAZINE, AQUEOUS SOLUTION  
 14.3. Transport hazard class(es) 8  
 Subsidiary Hazard Class 6.1  
 14.4. Packing group II

14.5. Environmental hazards Dangerous for the environment  
 Product is a marine pollutant according to the criteria set by IMDG/IMO

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component           | CAS No     | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|---------------------|------------|-----------|--------|-----|-------|------|----------|------|------|
| Hydrazine (hydrate) | 10217-52-4 | -         | -      | -   | X     | X    | -        | X    | X    |
| Hydrazine           | 302-01-2   | 206-114-9 | -      | -   | X     | X    | KE-19981 | X    | X    |
| Water               | 7732-18-5  | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |

| Component           | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|---------------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Hydrazine (hydrate) | 10217-52-4 | -    | -                                             | -   | -    | -    | X     | -     |
| Hydrazine           | 302-01-2   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Water               | 7732-18-5  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component           | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                  | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------|------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hydrazine (hydrate) | 10217-52-4 | -                                                                   | -                                                                                              | -                                                                                                     |
| Hydrazine           | 302-01-2   | -                                                                   | Use restricted. See item 28.<br>(see link for restriction details)<br>Use restricted. See item | SVHC Candidate list - 206-114-9 - Carcinogenic, Article 57a                                           |

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

|       |           |   |                                           |   |
|-------|-----------|---|-------------------------------------------|---|
|       |           |   | 75.<br>(see link for restriction details) |   |
| Water | 7732-18-5 | - | -                                         | - |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

## REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Seveso III Directive (2012/18/EC)

| Component           | CAS No     | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Major Accident<br>Notification | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Safety Report<br>Requirements |
|---------------------|------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Hydrazine (hydrate) | 10217-52-4 | Not applicable                                                                                  | Not applicable                                                                                 |
| Hydrazine           | 302-01-2   | 0.5 tonne                                                                                       | 2 tonne                                                                                        |
| Water               | 7732-18-5  | Not applicable                                                                                  | Not applicable                                                                                 |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 3 (self classification)

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                                                            |
|-----------|---------------------------------------|------------------------------------------------------------------------------------|
| Hydrazine | WGK3                                  | Krebserzeugende Stoffe - Class I : 0.05 mg/m <sup>3</sup><br>(Massenkonzentration) |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

## Full text of H-Statements referred to under sections 2 and 3

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

H226 - Flammable liquid and vapor  
H301 - Toxic if swallowed  
H311 - Toxic in contact with skin  
H330 - Fatal if inhaled  
H314 - Causes severe skin burns and eye damage  
H317 - May cause an allergic skin reaction  
H318 - Causes serious eye damage  
H350 - May cause cancer  
H400 - Very toxic to aquatic life  
H410 - Very toxic to aquatic life with long lasting effects

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Creation Date** 11-Sep-2014

**Revision Date** 27-Sep-2023

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and**

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

Revision Date 27-Sep-2023

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## Preparations).

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**